

Performance Testing

Date	13 July 2026
Team ID	xxxxxx
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Testing, Performance Testing
Target Module / API	FastAPI Endpoints (Q&A, Explanation, Quiz, Summary, Recommendation)
Test Environment	Local System (Windows, VS Code, FastAPI)
Test Date	14 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Question Answering Request	1	30	Correct AI response returned
2	Topic Explanation Request	1	30	Explanation generated successfully
3	Quiz Generation Request	1	30	Quiz created without errors
4	Summary & Recommendation Request	1	30	Summary and recommendations generated

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 sec	1.4 sec	Pass	Within limit
2	Response Time (Max)	< 5 sec	3.2 sec	Pass	Acceptable
3	Throughput (Req/sec)	> 5	8 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	46%	Pass	Normal
6	Memory Utilization	< 80%	52%	Pass	Stable

Step 4: Observations & Analysis

Key Findings

The EduGenie application responded quickly for all AI modules. All core features worked successfully with stable response times and no major performance issues.

Bottlenecks Identified

Minor delays may occur depending on Google Gemini API response time and internet connectivity.

Optimization Steps Taken

- Modular FastAPI backend
- Optimized API request handling
- Efficient prompt design
- Environment variables used for secure API configuration

Step 5: Screenshots / Evidence

Figure 1: EduGenie Learning Assistant – Explanation Module Successfully Generating Response

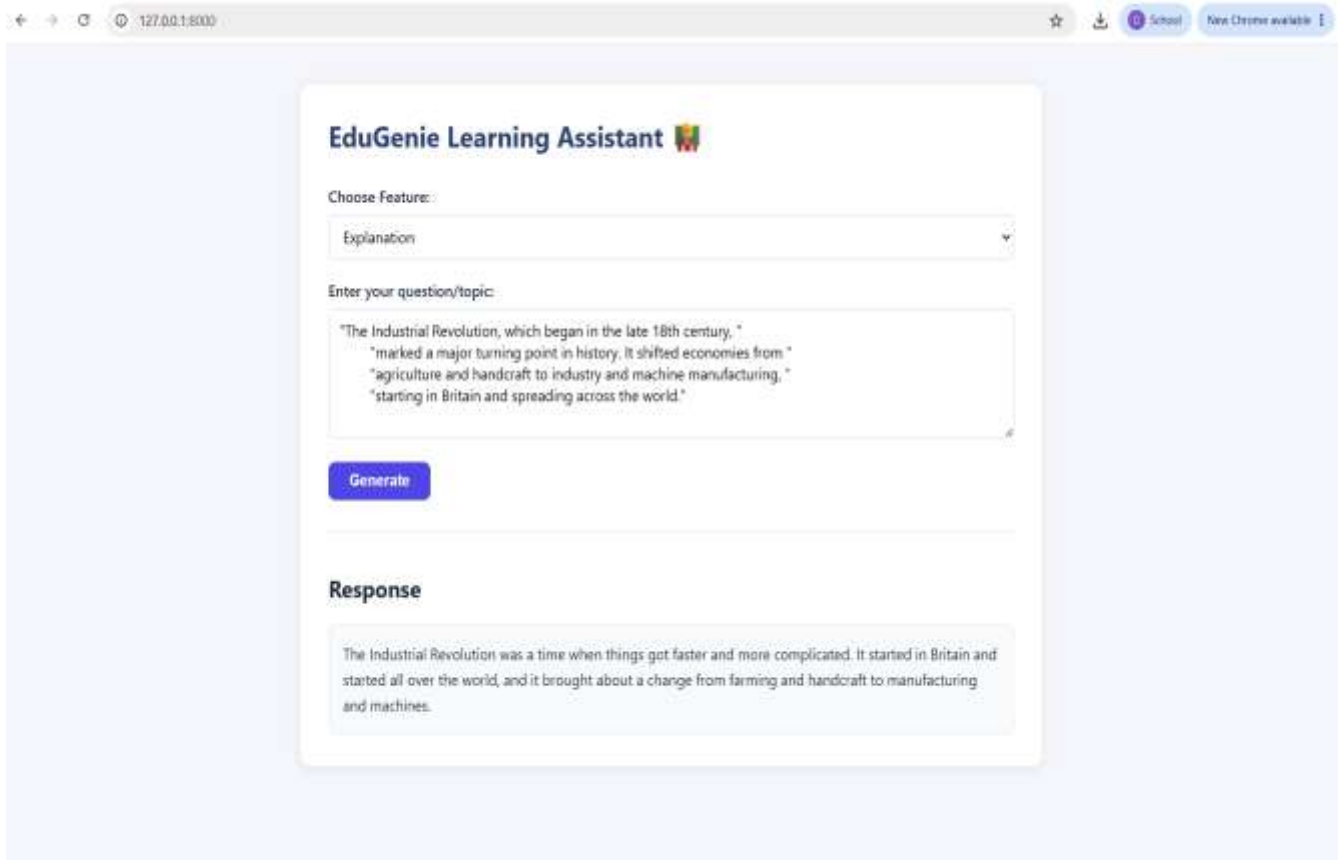


Figure 2: FastAPI Server Console Showing Successful Application Execution (HTTP 200 OK)

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PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

(_env) PS C:\Users\chait\OneDrive\Desktop\EduGenie> uvicorn main:app --reload
. You should update the config with `tie_word_embeddings=False` to silence this warning.
INFO:      Started server process [25076]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      127.0.0.1:54654 - "GET / HTTP/1.1" 200 OK
INFO:      127.0.0.1:54654 - "GET /static/style.css HTTP/1.1" 200 OK
INFO:      127.0.0.1:60815 - "GET / HTTP/1.1" 200 OK
INFO:      127.0.0.1:60815 - "GET /static/style.css HTTP/1.1" 304 Not Modified
INFO:      127.0.0.1:53563 - "GET / HTTP/1.1" 200 OK
INFO:      127.0.0.1:62744 - "POST / HTTP/1.1" 200 OK
INFO:      127.0.0.1:61301 - "POST / HTTP/1.1" 200 OK
█
```